

Infrastructure Mastery Manual: Vol 3

Technical Reference: Modules, 3-Tier Arch, & EKS

Section 1: Terraform Modules

Modules provide encapsulation. A module is simply a directory of HCL; by mapping outputs to inputs, you create an interface that hides underlying complexity.

```
# Parent Module -> Child Module Interface
module "vpc" {
  source      = "../modules/vpc"
  environment = "production"
  # Sibling output passing via Parent only
}
```

Section 2: 3-Tier Infrastructure Engineering

Using `locals` and `for_each` eliminates copy-paste configurations, allowing for scalable, multi-AZ deployment with centralized management of security and lifecycle policies.

```
# RDS Lifecycle Compliance
resource "aws_db_instance" "main" {
  engine           = "mysql"
  instance_class   = "db.t3.medium"
  # Prevent accidental deletion in Production
  lifecycle {
    prevent_destroy = true
  }
}
```

Section 3: EKS (Elastic Kubernetes Service)

EKS requires specific security and networking integrations to be production-ready. The AWS VPC CNI ensures pods possess native IP addressability within the VPC.

```
# Security Engineering: IRSA
# Map AWS IAM Roles to K8s Service Accounts (IRSA)
# Avoid node-level administrative rights.
# Use OIDC identity providers to bridge the trust between EKS and IAM.

# AWS Load Balancer Controller:
# Automates application ingress routing via ALBs/NLBs.
```